

* **Computer Security:** The protection afforded to an automated information system in order to attain the applicable objectives of preserving the integrity, availability and confidentiality of information system resources such as hardware, software, firmware, information/data and telecommunication.

This introduces three key objectives.

• **Confidentiality:** This term covers two related concepts.

1) **Data Confidentiality:**

Assures that private or confidential information is not made available or disclosed to unauthorized individuals.

2) **Privacy:**

Assures that individuals control or influence what information related to them — ^{who} may be collected and stored, use it and share it.

• **Integrity:** This term covers two related concepts.

1) **Data Integrity:** Assures that information and programs are changed only in a specified and authorized manner.

2) **System Integrity:**

Assures that system performs its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation of the system.

• **Availability:** Assures that systems work promptly and service is not denied to authorized users.

These three Concepts for what is often referred to as the "CIA Triad".

There are two more additional triads

* **Authenticity**: The property of being genuine and being able to be verified and trusted, Confidence in the validity of a transmission, a message, or message originator.

• It means verifying that users are who they say they are and that each input arriving at the System came from a trusted Source.

* **Accountability**: The Security goal that generates the requirement for actions of an entity to be traced uniquely to that entity.

• It means every action taken in a System can be traced back to the person (or System) who did it.

* **The OSI Security Architecture.**

• The OSI Security architecture is internationally recognized and provides a Standardized technique for deploying Security measures within an organization.

• It focuses on three major concepts:

1. Security Attacks.

2. Security Services.

3. Security Mechanism.

• The OSI model is based on divide and conquer, it splits up the communication system into 7 abstract layers.

• The Physical Layer

• The Data Link layer

• The Network Layer

• The transport Layer

• The Session Layer

• The Presentation Layer

• The application Layer.

• Security Attack: Any action that compromises the security of information owned by an organization

• Security Mechanism: A process (or device incorporating such a process) that is designed to detect, prevent, or recover from a security attack.

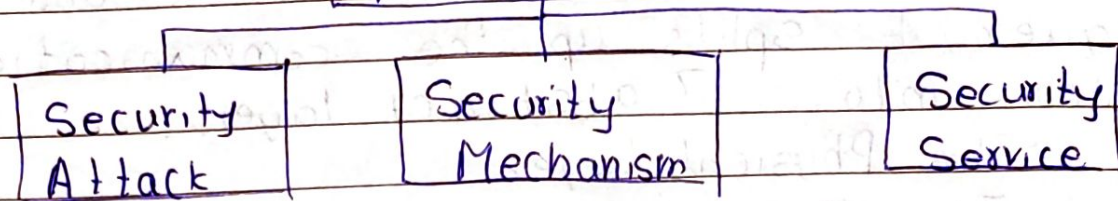
• Security Service: A processing or communication service that enhances the security of the data processing systems and the information transfers of an organization.

A service is intended to counter security attacks, and they make use of one or more security mechanisms to provide service.

* OSI Security

OSI (Open Systems Interconnection) Security refers to a set of protocols, standards, and techniques used to ensure the security of data and communications in a network environment.

OSI Security Architecture



* Security Attack:

A Security attack is an attempt by a person or entity to gain unauthorized access to disrupt or compromise the Security of a system, network, or device.

1. Passive Attack:

- Attacks in which a third part intruder tries to access the message / content / data being shared by the Sender and receiver by keeping a close watch on the transmission or eave-dropping the transmission.
- The goal of the attacker is to obtain information that is being transmitted.



- The attacker can only see the data. He/She will not modify it
- It is difficult to detect
- We can prevent it by using better encryption techniques.

- Two types of passive attacks are
 1. Release of Message Contents
 2. Traffic Analysis

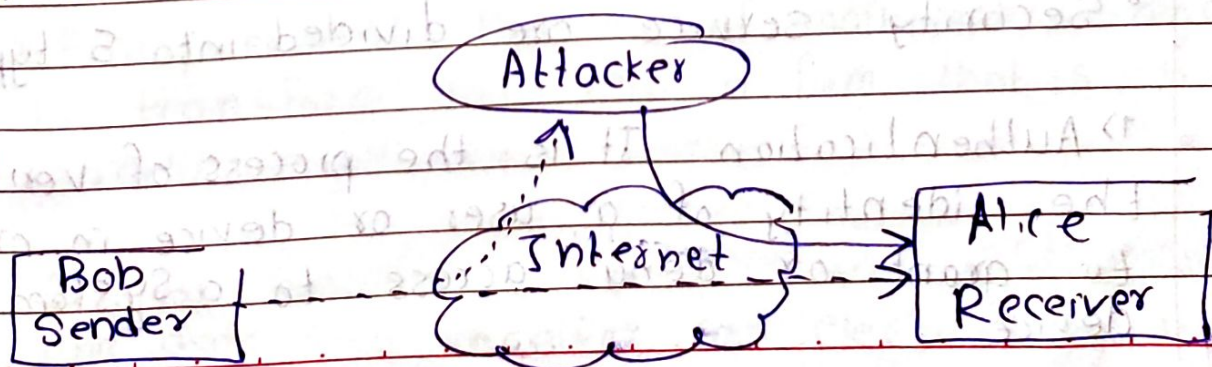
- In first case attacker intercepts and can read/listen to the actual message. It can be done using variety of techniques, such as packet sniffing or man in middle attacks.

- In traffic Analysis attacker analyses the network traffic pattern and metadata to gather information about system, network, or device.

- Here the intruder can't read the message but only understand the pattern and length of encryption.
- It can be performed using flow analysis, protocol analysis.

2. Active Attack :

- It involves attacker actively disrupting or altering system, network or device activity.
- The goal is to cause damage by modifying data rather than gathering information.



• Four types of active attacks are

1) Masquerade : In this the attacker pretends to be an authentic sender. It may involve attacker using stolen credentials

2. Modification of Message : Some portion of legitimate message is altered, or that message is delayed or reordered.

3 Replay : It involves passive capture of a data unit and retransmission of message through a passive channel.

4. Denial of Service (DOS) : Attacker Sends a large volume of data to overload the Server to degrade performance. It doesn't allow Server to fulfill legitimate requests.

* Security Service :

• Security Service refers to the different service available for maintaining the security and safety of an organization.

• They help in preventing any potential risks to security.

• Security service are divided into 5 types :

1) Authentication : It is the process of verifying the identity of a user or device in order to grant or deny access to a system or device

2> Access Control: It involves the use of policies and procedures to determine who is allowed to access specific resources within a system.

3> Data Confidentiality: Data Confidentiality is responsible for the protection of information from being accessed ~~specific resources~~ or disclosed to unauthorized parties.

4> Data Integrity: It is a security mechanism that involves use of techniques to ensure that data has not been tampered with or altered in any way during transmission or service.

5> Non-Repudiation: It makes sure that someone cannot deny having done something such as sending a message or making a transaction.

* Security Mechanism

- The mechanism that is built to identify any breach of security or attack on the organization, is called a security mechanism.
- It is also responsible for protecting a system, network, or device against unauthorized access, tampering or other security threats.

1> Encryption: The use of mathematical algorithms to transform data into a form that is not readily intelligible.

2> Digital Signature: Sender can electronically sign the data and receiver can electronically verify the signature to protect against forgery.

3) Data Integrity: It provides a check value by the sender to check whether the ~~measure~~ message is modified or not.

4) Authentication Exchange: Two entities exchange some message to prove their identity to each other.

ex. bluetooth, hotspot

5) Traffic Padding: Extra bits are added to actual data.

• The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.

6) Routing Control: Selecting and continuously changing different available routes to avoid eaves dropping on a particular route.

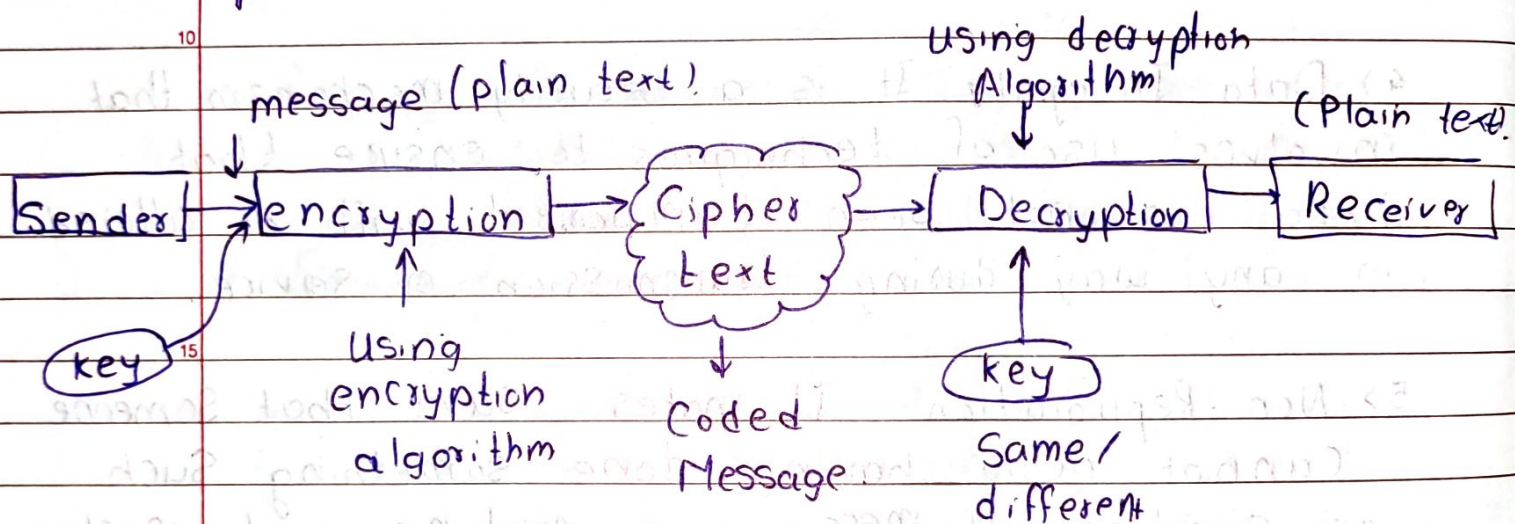
7) Access Control: It is a method to check whether a user has access right of the data.

8) Notarization: The use of trusted third party to control the communication between two entities.

This can be done to prevent repudiation.

* **Cryptography** : The art of protecting information by transforming it into unreadable format is cryptography.

• More generally cryptography is about constructing and analysing protocols that prevent third-parties or the public from reading private messages.



Encryption involves the use of an encryption algorithm, denoted by E , and an encryption key, e . Likewise, decryption involves use of a decryption algorithm, denoted by D , and a decryption key, d .

These operations are summarized below

$$C = E_e(P)$$

$$P = D_d(C)$$

where C = Cipher text
 P = plain text.

Here key e, d can be same or different.

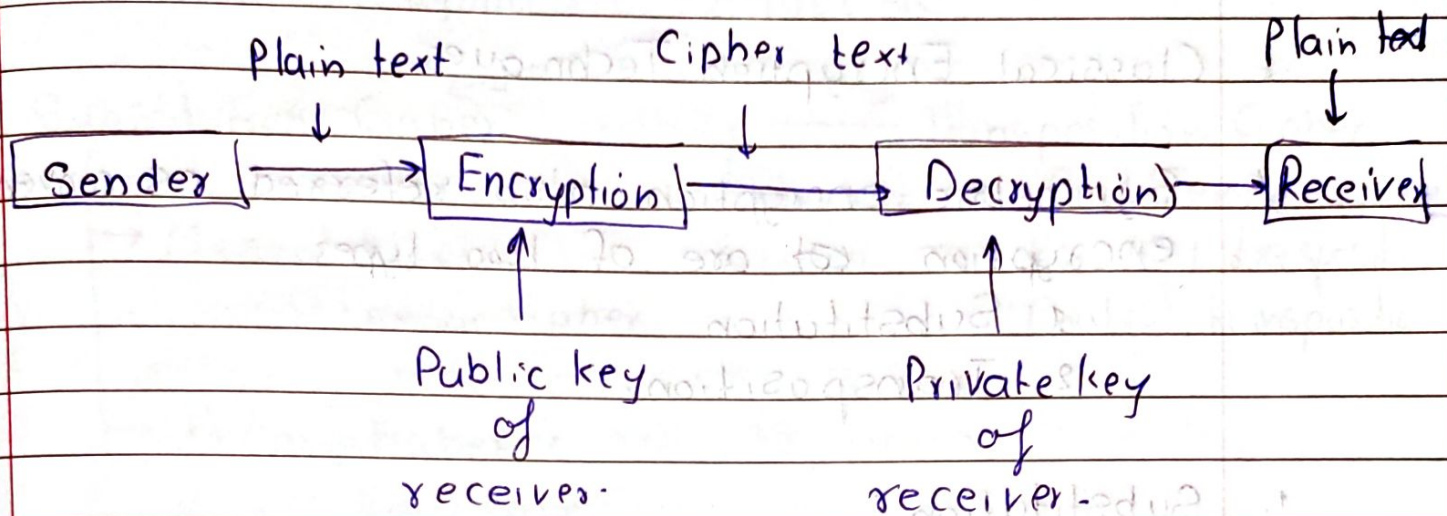
* Types of Cryptography.

1) Symmetric key Cryptography.

- Also known as Secret key Cryptography or private key cryptography.
- Same key is used for both encryption and decryption.
- Examples AES, DES.
- Symmetric key Cryptography is faster and Simple.
- The Sender and receiver must know (or exchange) same key somehow. (Both must have access to key).
- It is ideal for encrypting large amounts of data.
- Main challenge is securely distributing the key.

2) Asymmetric key Cryptography

- Also called as public key cryptography.
- It uses a pair of keys - a public key for encryption and a private key for decryption.
- It solves the key distribution problem.
- Sender encrypts the message using receiver's public key so that only receiver can decrypt it using its private key.
- Uses digital signatures for authentication.
- It is slower than symmetric.
- Ex. RSA, ECC, DSA. (Digital Signature Algorithm, Elliptic Curve Cryptography).

Symmetric key

- Also known as Conventional encryption technique or private key cryptography.

- Uses One key.
- Same key for both encryption and decryption.

- Key distribution is difficult (needs Secure Channel).

- Used for bulk data.

- No built-in authentication.

- Commonly used algorithms are AES, DES.

- Faster.

- Less Complex.

Diagram.

Asymmetric key

- Also known as public key cryptography.

- Uses two keys.
- Different key for both encryption and decryption.

- Key distribution is easier as only public key is shared openly.

- Used for exchange of Secret keys, Signatures.

- Built-in authentication using digital Signatures.

- Commonly used algorithms RSA, Diffie-Hellman, ECC, DSA.

- Slow.

- More Complex.

Diagram.

* Classical Encryption Techniques.

* Symmetric encryption also referred as conventional encryption are of two types.

1. Substitution
2. Transposition.

1. Substitution :

Substitution is one in which letter of plain text are replaced by another letter, numbers or symbols according to a rule or key.

Rule

ex A - P

B - 1

CAB → ZP1.

E - Z

D - @

F - \$

* Transposition :

• A transposition cipher is a type of encryption technique where the position of the letters in the plain text message are rearranged to form a ciphertext message.

• Some sort of permutation is performed on the plain text.

ex: Name → mane, aenm, mnea

Symmetric Encryption

Substitution Cipher

- Monoalphabetic
 - Caesar Cipher

→ Polyalphabetic

- Play Fair
- Hill Cipher
- One time pad
- Vigenere
- Vernam

Transposition Cipher

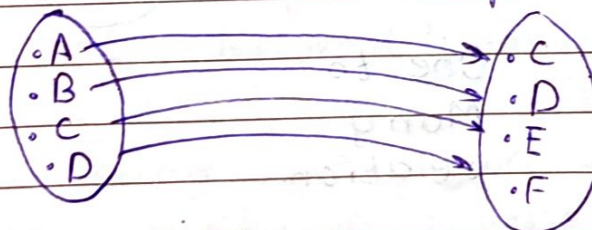
- Keyless → Rail Fence
- Keyed → Row/Col transposition
- Double transposition

* Monoalphabetic Substitution Cipher:

- It is one of the most fundamental classical encryption techniques, where each letter in the plaintext is replaced by exactly one unique letter from the cipher text alphabet.

Plain Text

Cipher Text



One
to
One
relation

- The relation between a character in a plain text to a symbol in cipher text is always one to one.

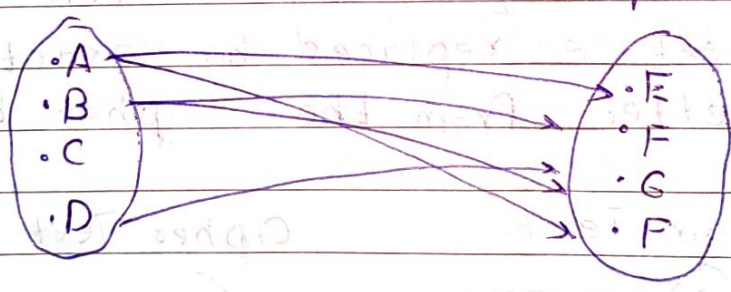
- fixed Substitution rules.
- The Cipher line can be a permutation of the 26 alphabetic characters.
- It can be cracked using frequency analysis.

* Polyalphabetic Substitution Cipher:

- It is a cipher technique where multiple cipher alphabets are used to encrypt the plain text.
- There is no fixed Substitution rules each occurrence of a character may have a different Substitute.
- The relationship between character in plain text to a character in cipher text is one to many.

Plain Text

Cipher Text



One to Many relation.

ex. $A : \{x, y\} \rightarrow$ based on some rule.
 $B : \{p, z\}$
 $C : \{x, s\}$

- Which cipher text character to be used is determined by a key.
- It is more secure and harder to crack using frequency analysis.

• Same letter in plain text can be encrypted to different Cipher text letter based on its position

• The main goal is to prevent pattern matching and to hide letter frequency.

1. Caesar Cipher

• It is one of the oldest and simplest method of encryption

• It is also called Shift Cipher or additive cipher.

• It is named after Julius Caesar, who used it to protect his military communications.

• This technique involves replacing a letter in plain text by a letter corresponding to the number of shifts.

• A mod by 26 is performed to keep letters between (A-Z)

$$C = (P + K) \bmod 26$$

$$P = (C - K + 26) \bmod 26$$

↑
No avoid Negative add 26.

• While decrypting we again shift the letter in cipher text in reverse direction

• If $(C - K)$ is negative while decryption we add 26.

key/Shift →							PT	CT (Shift 3)
X	Y	Z	A	B	C	D	A	D
⋮	⋮	⋮	⋮	⋮	⋮	⋮	B	E
⋮	⋮	⋮	⋮	⋮	⋮	⋮	C	F
⋮	⋮	⋮	⋮	⋮	⋮	⋮	D	G
A	B	C	D	E	F	G	...	...

- The key is in between $1 \leq k \leq 25$.
- Caesar Cipher is Monoalphabetic Substitution technique. So, it is easily predictable and can be easily cracked by brute-force.
- Easy to crack (Only 25 possible key/shift values)

★ Play Fair Cipher

- Play fair was the first practical digraph Substitution cipher. (Pair of letters encrypted together)
- It was invented in 1854 by Charles Wheatstone but was named after Lord Playfair who promoted the use of this cipher.
- In playfair we encrypt ~~to~~ pair of alphabets (digraphs) instead of single alphabets.
- It was used for tactical purpose by British forces in WWI.
- In this a 5×5 matrix of letters is constructed from a keyword.

Plain text = HELLO $\rightarrow$ digraph HE LX LO

- It is more secure than Caesar.
- It is vulnerable to digraph frequency analysis.
- It is fast as it encrypts a Pair of characters at a time

* Hill Cipher :

- The Hill Cipher is a polygraphic Substitution cipher based on matrix multiplication over modulo 26 arithmetic.
- It is developed by Lester S. Hill in 1929.
- It encrypts multiple letters at once using linear transformations (polygraphic : digraph, trigraph etc).

- A key matrix ($n \times n$) is used for encryption.
- This method makes use of following

$$* C = kP \text{ mod } 26$$

$$* P = k^{-1}C \text{ mod } 26.$$

- To encrypt each block of n letters is multiplied by a $n \times n$ matrix (invertible) called key.

- Every operation is performed mod 26.
- To decrypt the blocks of Cipher text are multiplied by inverse of key.

- It is faster because it is polygraphic.
- It is more complex to decrypt.
- It is strong against Frequency Analysis.
- It is Scalable (we can increase block size 2×2 , 3×3 etc).

* Vigenere Cipher:

- It is named after Blaise de Vigenere who is a French mathematician (16th Century).
- It is a polyalphabetic substitution cipher.
- Encryption is done using 26×26 matrix where alphabets are written out 26 times, each alphabet shifted cyclically the left (corresponding to 26 possible Caesar Ciphers).
- Here key is repeated to match the length of plain text
ex. key: HELLO

Plain Text : GIVEMEMONEY

respective key: HELLO HELLOH

encryption: For each character

- Find the row corresponding to key letter
- Find column corresponding to plain text letter
- The intersecting letter is cipher text letter.

Decryption: For each character:

- Find row of key letter
- Find column in that row where the cipher text letter appears.
- The column header is original plaintext letter.

It can also be solved mathematically using

$$C_i = (P_i + k_i) \bmod 26$$

$$P_i = (C_i - k_i + 26) \bmod 26$$

↑

* Vernam (One time Pad)

- The Vernam Cipher is a Symmetric key Cipher invented by Gilbert Vernam in 1917.
- It is a polyalphabetic Substitution Cipher.
- In this the number of characters in plain text and key are equal.

$$\text{Length (Key)} = \text{Length (Plain text)}$$

- It is similar in structure to the Vigenere Cipher, but with crucial differences in key quality and security.
- In Vigenere key is repeated which can be cracked using pattern evaluation but this problem is solved in Vernam.

$$C_i = (P_i + k_i) \bmod 26$$

$$P_i = (C_i - k_i + 26) \bmod 26$$

- It is a special case of Vernam Cipher where length of key matches length of plain text and key is completely random.

* Rail Fence:

- The Rail fence Cipher is keyless and a transposition cipher that writes the plaintext in a zig-zag (wave pattern) across multiple "rails" (rows).

- Plain text is written down as a sequence of diagonals and then read as a sequence of rows.
- In this type the existing characters in plain text are rearranged.

Encryption:

- Choose the number of rails (levels)
- Write the message in a zigzag pattern.
- Read the characters row by row

Decryption:

- Write the Cipher text row by row into the placeholders as per no of rails and length.
- Read characters in zigzag order
- It is easy to implement
- It is not very secure.
- Does not hide letter frequency.

* Row Column Transformation.

- This is a classical cipher that encrypts the message by writing it into a matrix (grid) row-wise, and reading it column-wise based on a key that determines column order.

Encryption Steps:

1. Choose a key (usually a word or number sequence)
2. Assign a number order to the letters of the key
3. Write the plain text into rows under columns labeled by the key.
4. Rearrange and read the columns based on the ordered key to get the cipher text

Decryption

- Use the key to determine column order
 - Create an empty grid with same dimensions
 - Fill the grid column-wise in key order using the cipher text
 - Read message row-wise to get the plaintext
- It is more secure than keyless (rail fence)
 - Still vulnerable to pattern analysis

* Keyed Transposition

- A keyed transposition cipher is a type of transposition cipher where the order of columns is determined by a keyword or key number.
- Here we divide plain text into groups of predefined (predetermined) size (blocks) and then use a key to permute the characters in each block separately

* Brute Force Attack.

- A brute force attack is one of the simplest (but often slowest) methods used to break a password, encryption key, or authentication system by trying every possible combination.
- The attacker systematically enumerates all possible keys or passwords in the search space.
- It is just raw trial and error.
- It involves finding all keys and trying all possible key to find the encrypted message.
- Software tools that can perform brute force attack are aircrack-ng, Davegrah, John the Ripper, hashcat, Hydra etc.
- If a password/key length is n and each position can have m possible symbols, the total combinations are

$$m^n$$

ex: 8-Character password using lowercase ($m=26$)

$$26^8 \approx 2.08 \times 10^{11} \text{ (208 billion combinations)}$$

Process:

Generate a key/Password

Test: (Try decryption / or login)

Repeat: Until Success

- It requires high processing power.
- It can be easily detected with rate-limiting and intrusion detection.

* Counter measure

- Use long and complex passwords (increases Search Space).
- Enable account lockout after failed attempts.
- Add Captcha to Slow down automated guessing.
- Use multifactor authentication.
- Employ key stretching.

• One of the way to stop brute force attack is captcha to whether human being is attacking on a system or not.

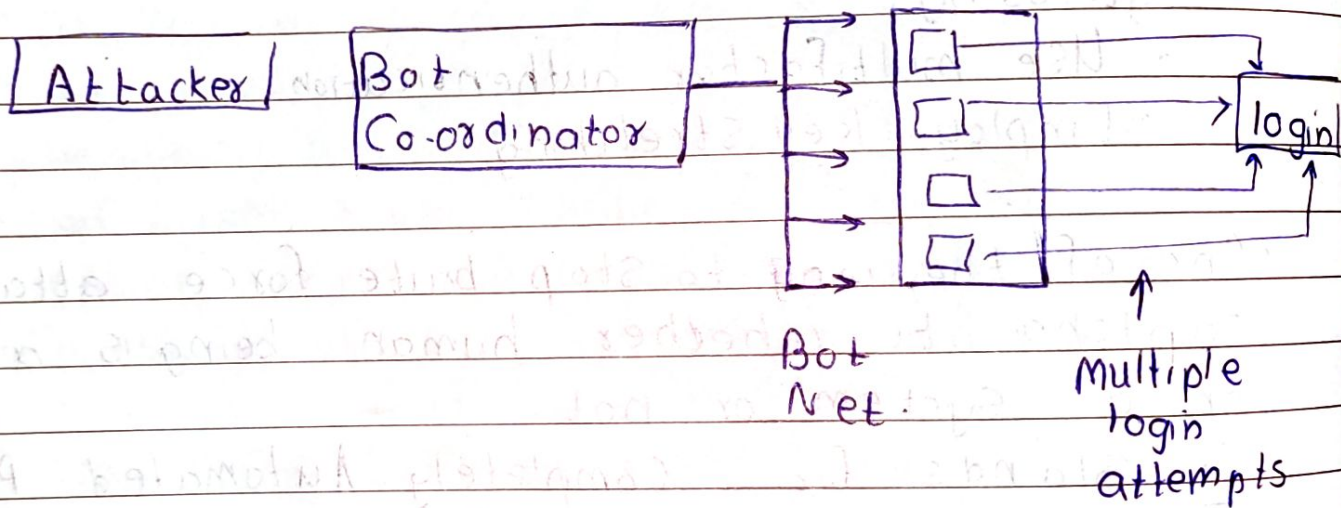
• It stands for Completely Automated Public Turing test to tell Computers and Humans Apart.

• The server presents a challenge (image, text, puzzle or interaction). The user solves it to prove it is human.

• Types of Captcha

- Text-based Captcha : Distorted characters/numbers in an image.
- Image based Captch : Identify objects in a set of images.
- Audio Captcha : listen to Spoken letters/number (for visually impaired)

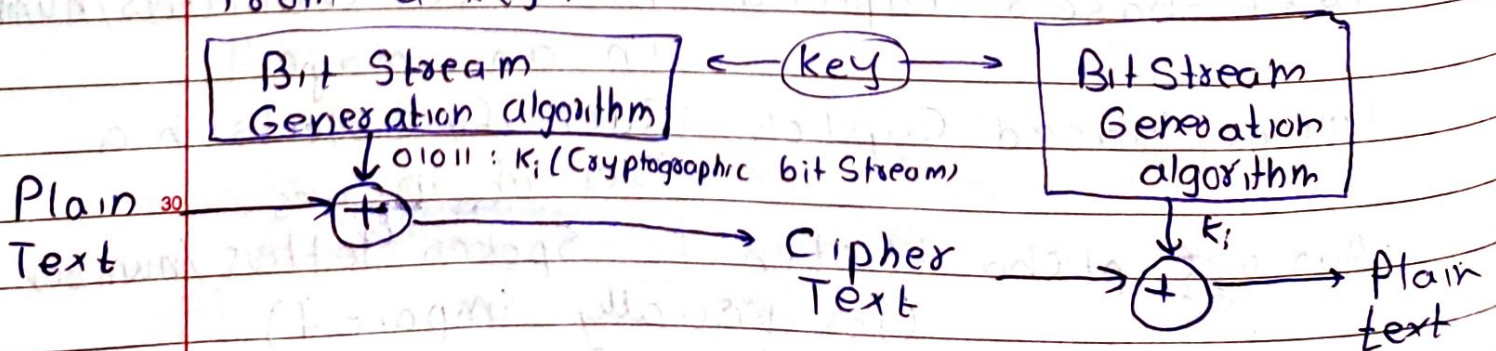
- Mathematical Captcha : Solve a Simple arithmetic problem.
- reCaptcha : "I'm not a robot" checkbox.
- 3D Captcha.
- Behavioral Captcha : Tracks mouse movements or keystroke patterns.



★ Stream & Block Cipher.

1. Stream Cipher :

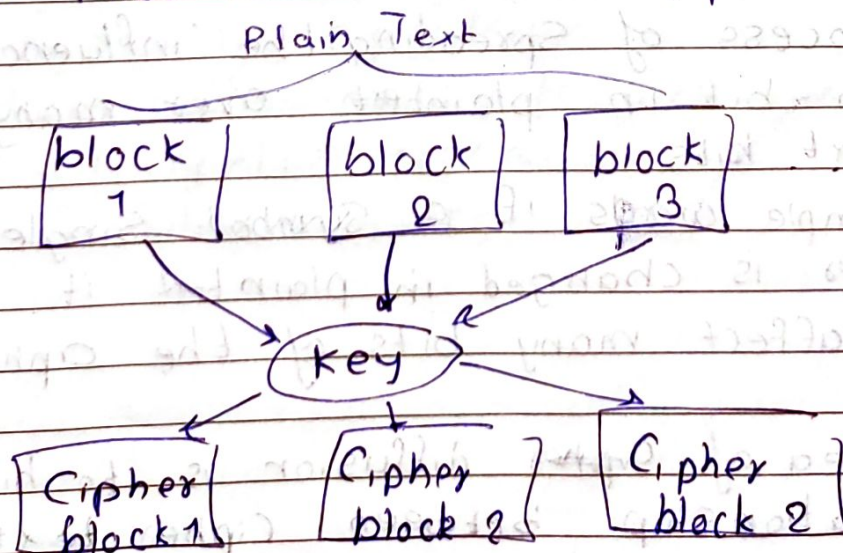
- It encrypts data one bit or byte at a time using a keyStream generated from a key.



- A pseudorandom key stream is generated from the Secret key.
- Plain text bit/bytes are XORed with the key stream bit/s/bytes to produce cipher text.
- For decryption same process (XOR CT with same bit stream).
- Used for fast real-time communication (audio/video).
- It can encrypt immediately without waiting for a full block.

2. Block Cipher:

- In this a block of plain text is treated as a whole and used to produce the cipher text of equal length.
- Typically a block size of 64 bit and 128 bits are used.
- Each block is encrypted using same key.
- If last block is shorter, padding is added.



* Shanon's Theory of Confusion and Diffusion

- It was introduced by Claude Shanon.
- Shanon's concern was to prevent cryptanalysis based on Statistical analysis.
- His idea was that A secure cipher must hide the relationship between the plain text, ciphertext, and key.

- If we assume that attacker has some knowledge of Statistical characters of plain text.

If this Statistics are in any way reflected in the cipher text.

- The cryptanalyst or attacker may be able to find the encrypted key.

Thus, there are two methods

1) Diffusion

2) Confusion

1. Diffusion :

- The process of spreading the influence of each bit in plaintext over many ciphertext bits.
- In simple words if a symbol single character is changed in plaintext it should affect many bits of the cipher text.
- The idea of ~~crypto~~ diffusion is to hide the relationship between ciphertext and plain text.

- It is achieved by permutation as well as transposition.
- It hides statistical structure of plaintext (eg letter frequency).
- ex. Shift rows in AES.

2. Confusion :

- It hides the relationship between ciphertext and the key.
- It makes the relationship as complicated as possible.
- If a single bit in a key is changed then most or all bits of the ciphertext will also change.
- Confusion means each bit of ciphertext should depend on several parts of key to make the connection between key and ciphertext unclear or difficult to understand.
- It is achieved by substitution operations.
- It changes data in a non-linear way, so small changes in the key produces large, unpredictable changes in ciphertext.
- Reduces the ability to reverse-engineer the encryption process.
- ex. SubBytes in AES.

★ Feistel Cipher Structure :

- It is a design model used to build many block ciphers.
- Most of the block cipher techniques follow this structure.
- Ex. DES.

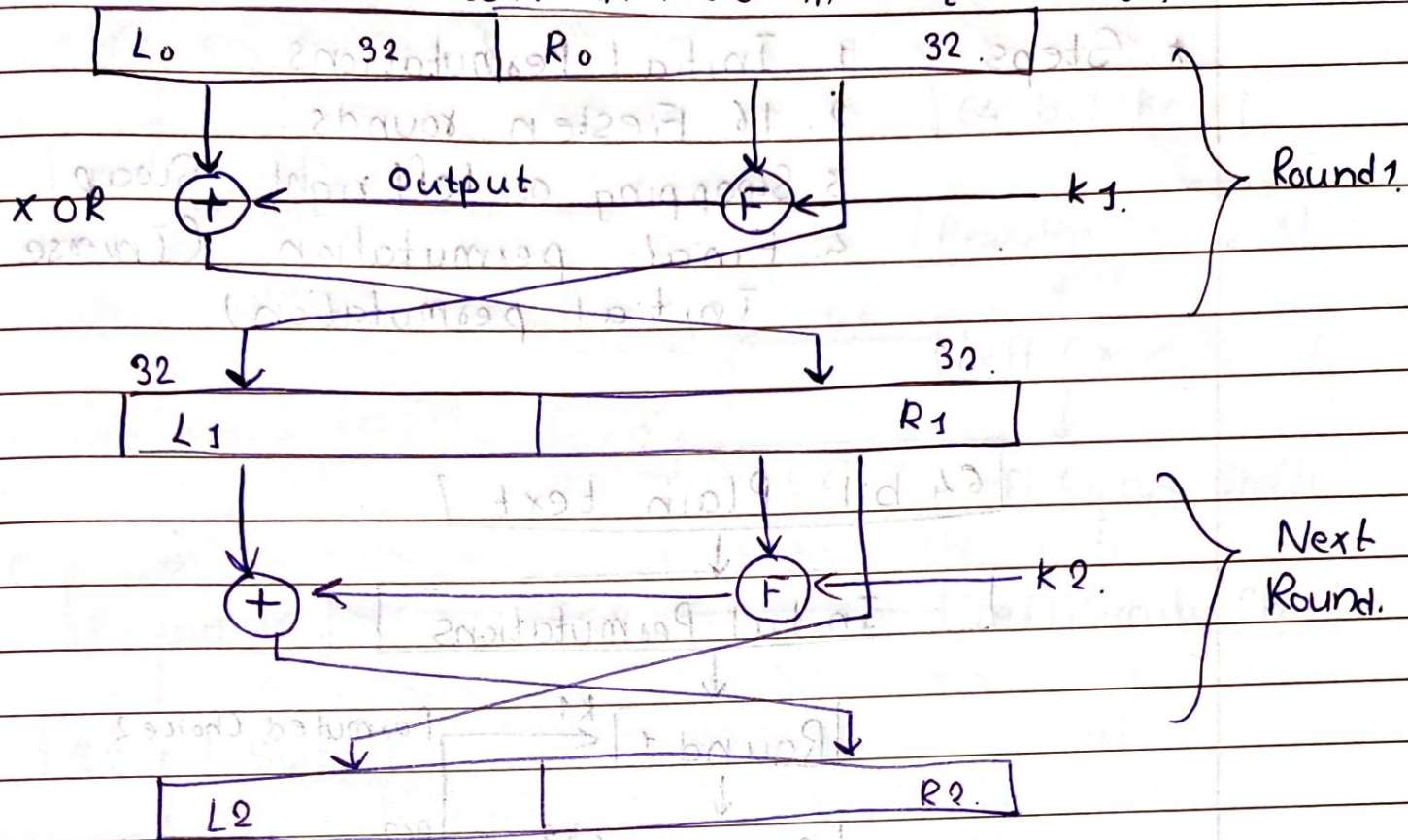
★ Steps :

- 1) The plain text is divided into two equal parts called as L_0 and R_0 .
- 2) The two halves of the data pass through N rounds of processing and then combine to produce the cipher text block.
- 3) On the right half we apply a function f and then we use subkey generated from master key.
- 4) The output of this is XORed with the left half and then this output will be swapped.

This is one single round, we will have N rounds depending on the algorithm and all rounds will have the same structure.

• In this algorithm we divide the plain text in two equal parts and apply the function on RHS and XOR it with LHS and the output is swapped then that algorithm follows Feistel structure.

Plain text divided in 2 equal Parts



* DES:

- It Stands for Data Encryption Standard (DES)
- It is a block Cipher technique
- It is Symmetric Cipher technique.
- (It is Symmetric key block Cipher technique)
- It has 64 bit plain text blocks.
- It encrypts the data in block of Size 64 bits each.
- It is based on the Feistel Cipher Model.
- Here, 16 rounds are calculated, where each round uses Feistel Structure.
- It was developed by IBM in 1970s
- Each round here uses 48-bit Subkey
- This 48-bit subkey is generated from the 56-bit effective key.

- * Steps :
1. Initial Permutations
 2. 16 Feistel rounds.
 3. Swapping or left right Swap
 4. Final permutation (Inverse Initial permutation)

64 bit Plain text

Initial Permutations

Round 1

Round 2

Round 16.

Permuted Choice 2

PC2

56 bit key

Initial 64 bit key. (Master key)

Final Permutation / Inverse Initial Permutation

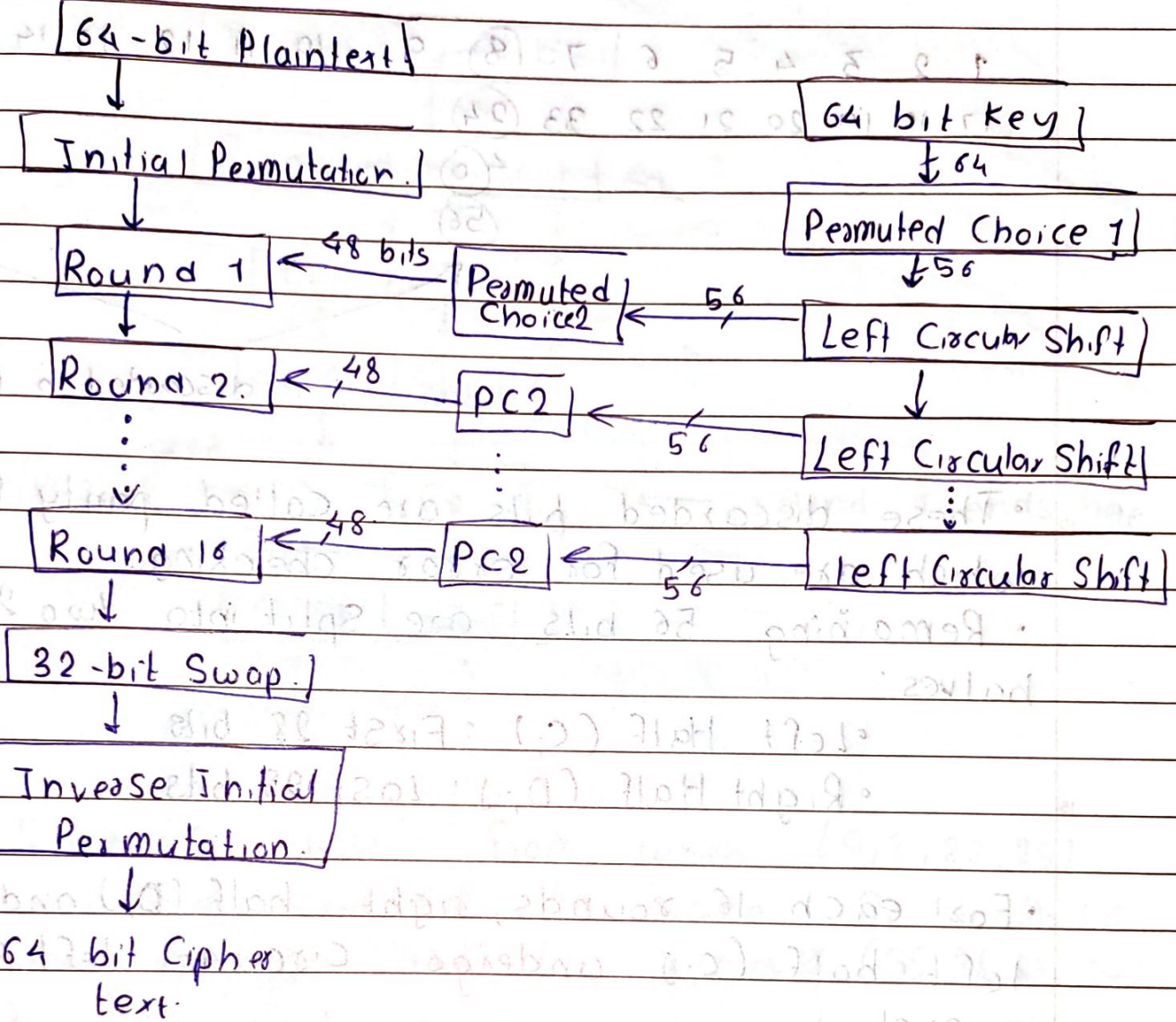
64 bit Cipher text.

* Initial Permutation

The 64 bit plain text is input into a Initial permutation (IP) function that rearranges the order of bit

The order of bits is changed using predefined table.

The IP table is a 8×8 matrix where each entry specifies the new position of a bit.



* Key Transformation :

- The 64 bit initial key is converted to 56 bit effective key.
- This 56 bit key further generates 48-bit Subkeys for each of 16 rounds.
- Initial 64 bit key first go through permuted choice 1 (PC-1) which reduces the key to 56 bits.
- In PC-1 every 8th bit is discarded. That is position 8, 16, 24, 32, 40, 48, 56 and 64 are discarded.

1 2 3 4 5 6 7 (8) 9 10 11 12 13 14 15 (Pr)
 17 18 19 20 21 22 23 (24)
 40
 56

discarded in PC-1

- These discarded bits are called parity bits which are used for error checking.
- Remaining 56 bits are split into two 28-bit halves:

- Left Half (C_i): First 28 bits

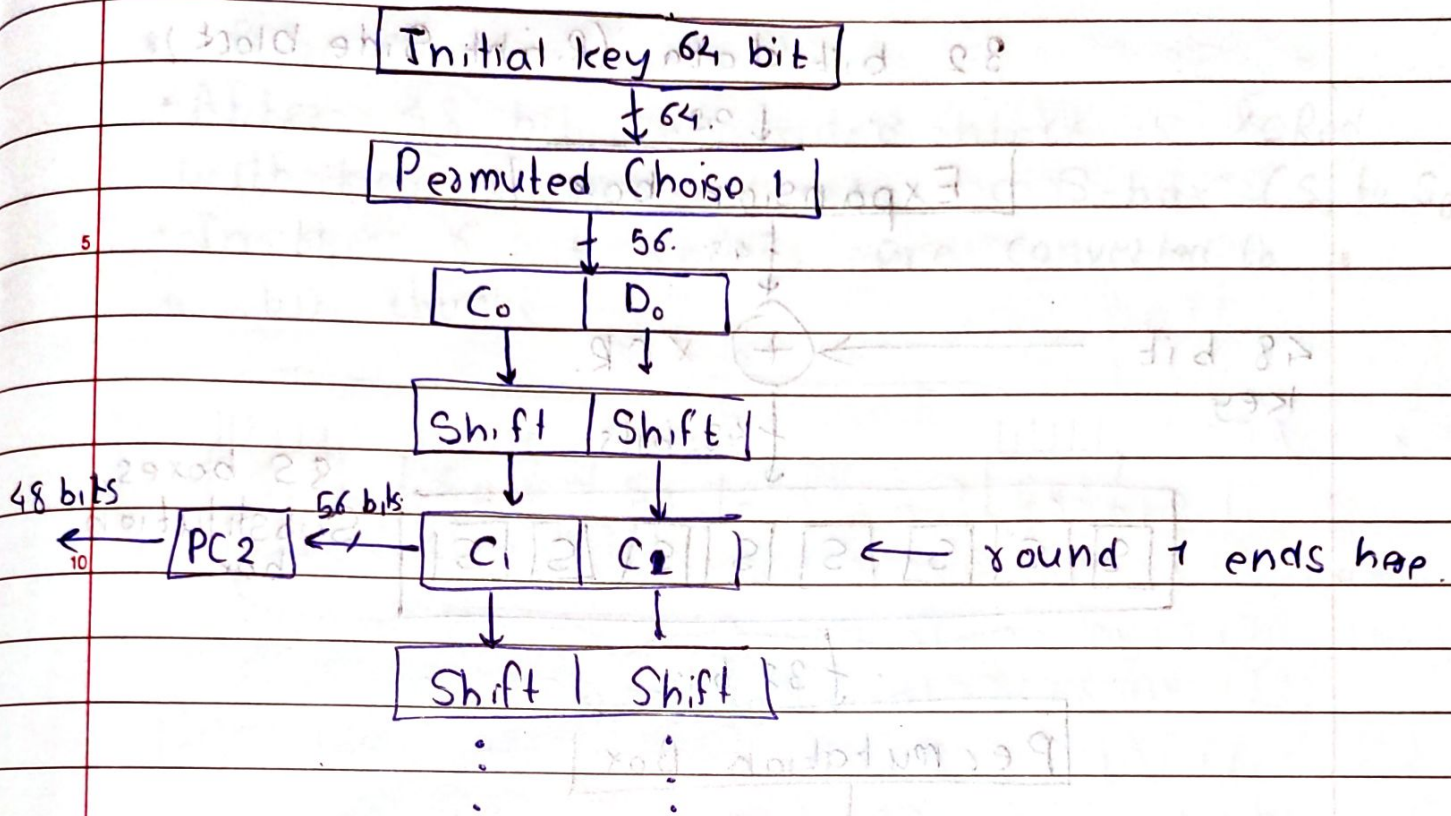
- Right Half (D_i): Last 28 bits

- For each 16 rounds, right half (D_i) and left half (C_i) undergo circular left shift operation.

- In round 1, 2, 9, and 16 both halves undergo 1-bit left shift operation. For other rounds the halves undergo 2-bit left shift operation.

- After circular shift operation C_i and D_i are combined into 56-bit block and then goes to PC2.

- Inside PC2 56 bits are converted to 48 bits using predefined table.



- C_1 has 28 bits from which (9, 8, 22, 25) bits are removed to make C_1 length 26
- Similarly Right half D_1 removes (35, 38, 33, 54) position bits.
- Thus we obtain 24+24 (48) bit key k_1

* How Function F_n works:

- The right side block of p_0 is processed using Mangler (F) function.
- It involves expansion, key mixing, Substitution (S-boxes) and permutation (P-box)

32 bit data (Right Side block)

↓ 32.

Expansion box

↓ 48

48 bit
key

⊕ XOR.

↓ 48 bits



8 S boxes.
Substitution
box.

↓ 32 bit

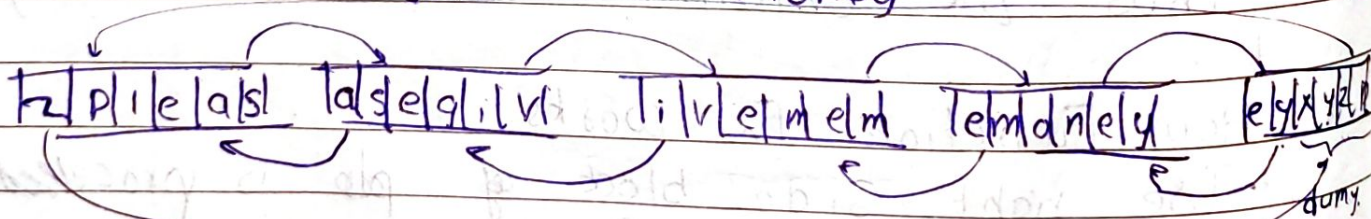
Permutation Box



32 bit key.

* Expansion Box

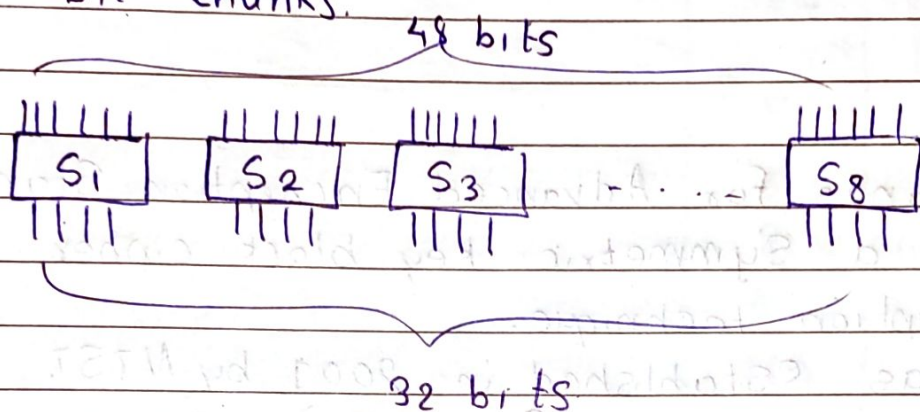
Please give me money



In this permutation 32-bit right block is converted expanded into 48 bits using expansion box or E-box table.

* S boxes.

- After 48 bit expanded block is XORed with key it is fed into a S-box (S_1 to S_8)
- In this 6 bit chunks are converted to 4 bit chunks.



- S-boxes are predefined lookup tables which reduces 6-bit chunk into 4 bit

ex. $001011 \rightarrow 001011$ $01 \rightarrow 1$ (row)
 $0101 \rightarrow 5$ (col)

	0	1	2	3	4	5	6	...
0	3	5	7	0	1	4	2	
1	4	2	1	0	9	7		

$7 = \underline{0111} \rightarrow 4$ bit chunk

We divide the 5 bit chunk in 2 parts 2 bit, 4 bit. Size. First and last bit are combined to form 2-bit part and rest make 4-bit

- We look for row and col in S_1 table based on these two parts.

* 32-bit Swap (final permutation,

- After 16 rounds we get two block 32 bit each these are Swapped again using inverse IP table (inverse of Initial IP).

* AES

- It Stands for Advanced Encryption Standard
- It is a Symmetric key block cipher encryption technique.
- It was established in 2001 by NIST.
- It operates on fixed size blocks of data of 128 bits i.e 16 bytes or 4 words.
- It replaces the older DES because DES was insecure due to its small key size.
- It has varying key size as per below

Rounds	Key Size	
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10	128 -bits	AES 128 Version
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12	192 bits	AES 192 Version
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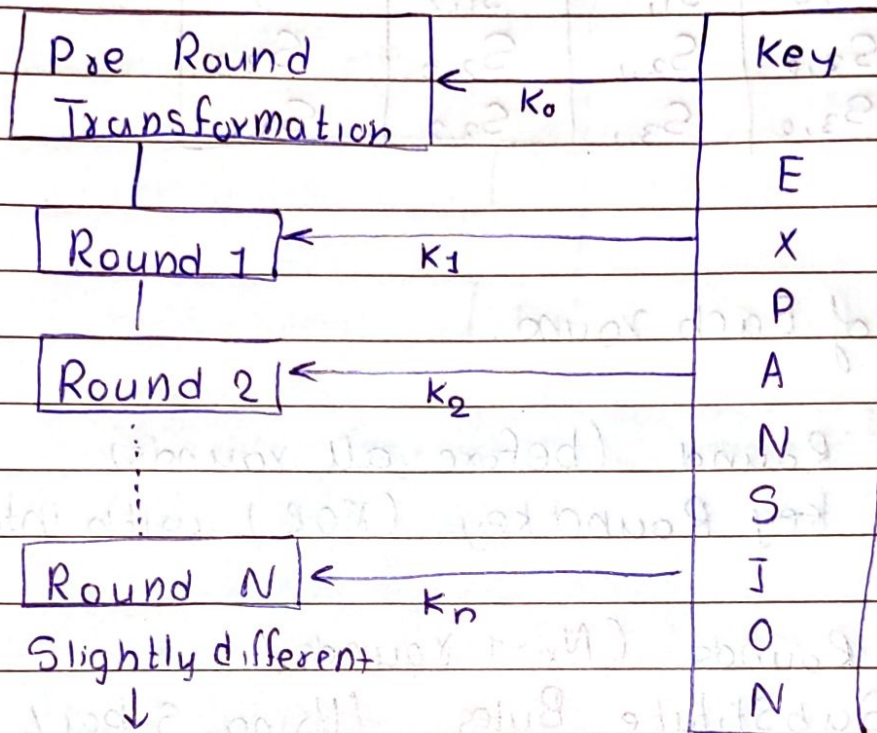
14	256 bits	AES 256 Version
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- AES transforms plain text into ciphertext through rounds of Substitutions and permutations
- Each round uses a different round key derived from the original key (Via the key expansion algorithm)

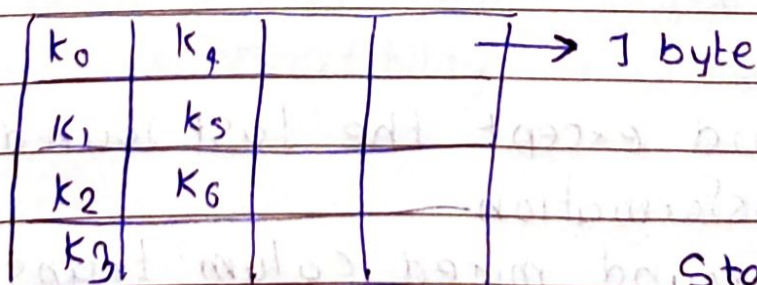
No of keys generated by key expansion algorithm equals to no. of round + 1

* General design of AES encryption

128 - bits Plain text



- In AES, all operations are performed on a two dimensional array of bytes called the State.
- It is organized as a 4×4 matrix.



State Stores intermediate results.

↓
1 word.

State Array.

$S_{0,0}$	$S_{0,1}$	$S_{0,2}$	$S_{0,3}$
$S_{1,0}$	$S_{1,1}$	$S_{1,2}$	$S_{1,3}$
$S_{2,0}$	$S_{2,1}$	$S_{2,2}$	$S_{2,3}$
$S_{3,0}$	$S_{3,1}$	$S_{3,2}$	$S_{3,3}$

★ 10 Structure of Each round.

1. Initial Round (before all rounds)

• Add key Round key (XOR) with initial key.

15 2. Main Rounds. ($N_r - 1$ rounds)

1. SubStitute Bytes (Using S box)

2. Shift Row

3. Mix Columns

4. Add round key.

20 3. Final Round (Nth round)

1. SubStitute Bytes

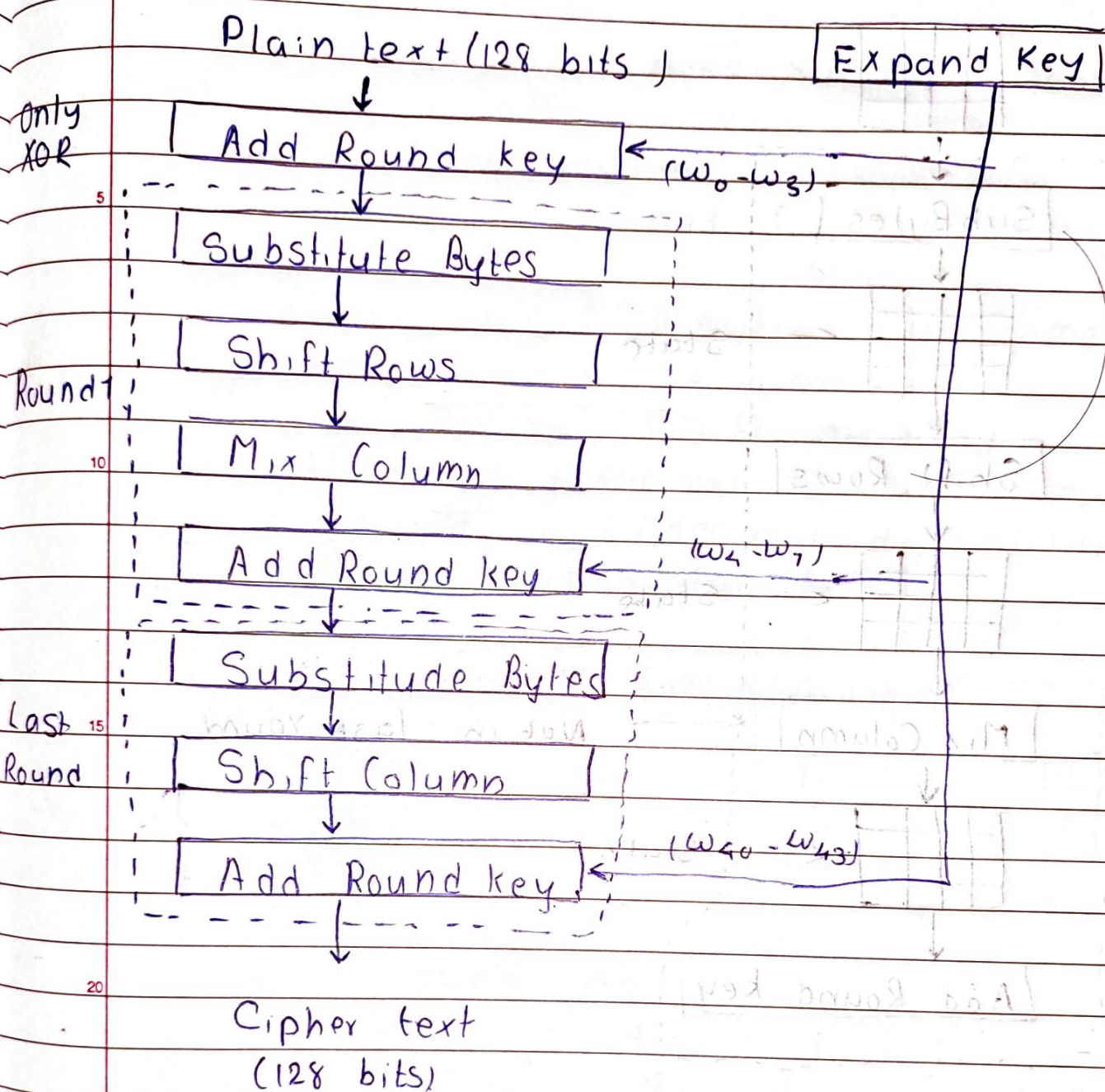
2. Shift Row

3. Add round key

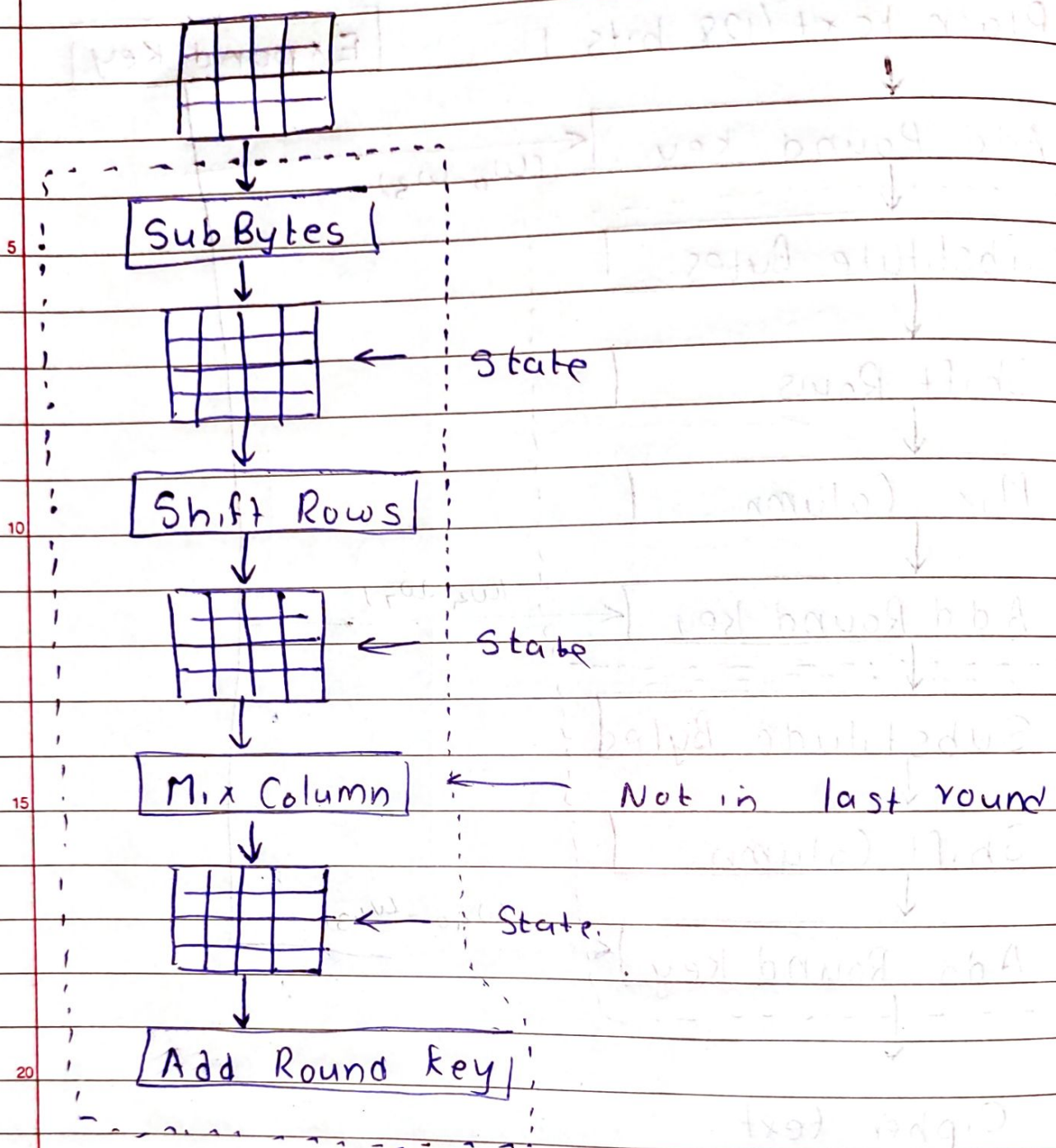
(No MixColumns Here)

25 • Each round except the last round uses four transformation.

30 • In last round mixed column transformation is missing



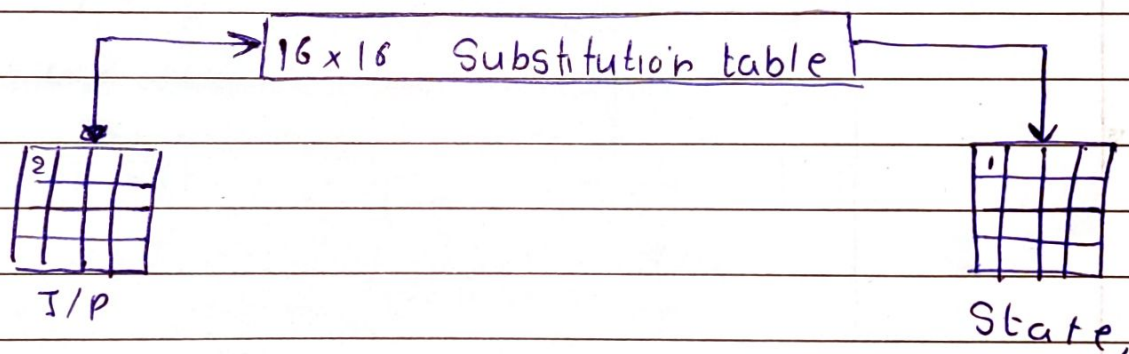
MixColumns is not applied in the final round to preserve invertibility and efficiency in decryption.



* There are four types of transformation

1. Substitution
2. Permutation
3. Mixing
4. Key Adding

- Like DES, AES also does Substitution but mechanism is different.
- Substitution here is done on each byte
- Only one table is used for transformation of bits
- This means that if 2 bytes are same then transformation is also same
- Here hexadecimal digits are used.
 - First hexadecimal digit determines row
 - Second hexadecimal digit determines column of the substitution table.



2 → 0000 0010

↓ ↓
 0 2
 (row) (column)

	0	1	2	3	4	...	16
0	1	2	(1)	3			
1							
2							
3							
⋮							